

On the Theory of Modules

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Let $A_1, \dots, A_s, \Gamma, \Delta$ be arbitrary classes of degree n , and let

$$\mathfrak{a}_1^{(i)}, \mathfrak{a}_2^{(i)}, \dots, \mathfrak{a}_{\psi(A_i)}^{(i)} \quad (i = 1, 2, \dots, s)$$

be all the modules of A_i . The number of systems of s modules

$$\mathfrak{a}_{q_1}^{(1)}, \mathfrak{a}_{q_2}^{(2)}, \dots, \mathfrak{a}_{q_s}^{(s)}, \quad (1)$$

which respectively belong to the classes $A_1, A_2, \dots, A_s$ is

$$\psi(A_1)\psi(A_2)\cdots\psi(A_s).$$

The number of systems (1) for which the least common multiple

$$[\mathfrak{a}_{q_1}^{(1)}, \mathfrak{a}_{q_2}^{(2)}, \dots, \mathfrak{a}_{q_s}^{(s)}] \quad (2)$$

is equal to a specified module of a class Θ is represented by the function $\bar{\gamma}(\Theta; A_1, \dots, A_s)$. Hence, if $\mathfrak{c}$ is any module of Γ , the number of systems (1) for which (2) becomes some divisor of $\mathfrak{c}$ belonging to the class Θ is

$$t(\Gamma \mid \Theta)\bar{\gamma}(\Theta; A_1, \dots, A_s).$$

Finally, to obtain the number of all systems (1) for which (2) is some divisor of $\mathfrak{c}$, one has only to form the sum

$$\sum_{\Theta} t(\Gamma \mid \Theta)\bar{\gamma}(\Theta; A_1, \dots, A_s),$$

where Θ runs through all classes of degree n . But the module (2) is a divisor of $\mathfrak{c}$ if and only if all modules in the system (1) are divisors of $\mathfrak{c}$. Therefore this same number is also represented by

$$t(\Gamma \mid A_1)t(\Gamma \mid A_2)\cdots t(\Gamma \mid A_s).$$

We therefore obtain the relation

$$\text{I.} \quad \sum_{\Theta} t(\Gamma \mid \Theta)\bar{\gamma}(\Theta; A_1, \dots, A_s) = t(\Gamma \mid A_1)t(\Gamma \mid A_2)\cdots t(\Gamma \mid A_s).$$

This relation, valid for every system of classes $\Gamma, A_1, \dots, A_s$, is fully sufficient to define the function $\bar{\gamma}$.

First, I shows that $\bar{\gamma}(\Gamma; A_1, \dots, A_s)$ vanishes if Γ is not a multiple of each of the classes $A_1, \dots, A_s$. In that case the product on the right-hand side of I vanishes. If we assume the assertion already proved for all divisors of Γ except Γ itself, then for every $\Theta \neq \Gamma$ one factor of the product under the summation sign vanishes, while $t(\Gamma \mid \Gamma) = 1$. Thus

$$\bar{\gamma}(\Gamma; A_1, \dots, A_s) = 0.$$

It follows that, for every class Γ , relation I can be given the form

$$\sum_{\substack{(\Theta, \Gamma) = \Theta \\ (\Theta, \Gamma') = \Gamma'}} t(\Gamma \mid \Theta)\bar{\gamma}(\Theta; A_1, \dots, A_s) = t(\Gamma \mid A_1)\cdots t(\Gamma \mid A_s), \quad (3)$$

where Γ' denotes the least common multiple of the classes $A_1, \dots, A_s$, and the summation extends over all classes Θ lying between Γ and Γ' , including the endpoints. If $\Gamma = \Gamma'$, this gives simply

$$\bar{\gamma}(\Gamma'; A_1, \dots, A_s) = t(\Gamma' \mid A_1)\cdots t(\Gamma' \mid A_s).$$

If, however, Γ is a multiple of Γ' , but distinct from Γ' , we suppose that before computing $\bar{\gamma}(\Gamma; A_1, \dots, A_s)$ we have computed the finitely many expressions $\bar{\gamma}(\Theta; A_1, \dots, A_s)$ in which Θ lies between Γ and Γ' and is different from Γ . Then (3) gives

$$\text{Ia. } \bar{\gamma}(\Gamma; A_1, \dots, A_s) = t(\Gamma | A_1) \cdots t(\Gamma | A_s) - \sum_{\substack{(\Theta, \Gamma) = \Theta \\ (\Theta, \Gamma') = \Gamma' \\ \Theta \neq \Gamma}} t(\Gamma | \Theta) \bar{\gamma}(\Theta; A_1, \dots, A_s),$$

and only known quantities occur on the right-hand side.

In the same way one obtains for $\bar{\delta}$, putting

$$\Delta' = (A_1, \dots, A_s),$$

the formulae

$$\text{II. } \sum_{\Theta} v(\Theta | \Delta) \bar{\delta}(\Theta; A_1, \dots, A_s) = v(A_1 | \Delta) \cdots v(A_s | \Delta),$$

$$\text{IIa. } \bar{\delta}(\Delta; A_1, \dots, A_s) = v(A_1 | \Delta) \cdots v(A_s | \Delta) - \sum_{\substack{(\Theta, \Delta) = \Delta \\ (\Theta, \Delta') = \Theta \\ \Theta \neq \Delta}} v(\Theta | \Delta) \bar{\delta}(\Theta; A_1, \dots, A_s).$$

The recursion formulae Ia and IIa show that the functions $\bar{\gamma}$ and $\bar{\delta}$ can be put together from t -functions and v -functions respectively by addition, subtraction, and multiplication. This representation is, however, substantially more complicated than the representation of t and v by ψ -functions. In the latter representation the number of representing ψ -functions is bounded by the degree n of the classes under consideration, whereas the number of t - and v -functions used in representing $\bar{\gamma}(\Gamma; A_1, \dots, A_s)$ and $\bar{\delta}(\Delta; A_1, \dots, A_s)$ is finite in each individual case but has no upper bound even when n and s are fixed. Moreover the subtraction occurring in these formulae makes the question of the vanishing of the $\bar{\gamma}$ - and $\bar{\delta}$ -functions very difficult. These difficulties are removed by transforming the expressions obtained for $\bar{\gamma}$ and $\bar{\delta}$ into products of functions which are easier to survey.

§ 10. Preliminary considerations.

1. From now on we restrict ourselves to primary classes. Such a class

$$A = \text{Cl}(p^{r_1} | p^{r_2} | \cdots | p^{r_n})$$

is completely determined by the system of exponents

$$r_1, r_2, \dots, r_n \tag{1}$$

and the prime number p . If only the exponent system (1) is given, we are dealing with a so-called partially determined primary class. In what follows, lower-case Greek letters always denote partially determined primary classes. If α is such a class and (1) is its system of exponents, we also write

$$\alpha = \text{Cl}(r_1 | r_2 | \cdots | r_n).$$

If, as in § 6,

$$r_1 = \cdots = r_{s_1} = e_1, \quad r_{s_1+1} = \cdots = r_{s_1+s_2} = e_2, \quad \dots, r_{n-s_k+1} = \cdots = r_n = e_k, \quad e_1 < e_2 < \cdots < e_k,$$

then (§ 6, III; a, b) gives

$$\begin{aligned}\psi(\alpha) &= p^{\sum_{i>h} s_i s_h (e_i - e_h)} \frac{\vartheta(p; n)}{\vartheta(p; s_1) \vartheta(p; s_2) \cdots \vartheta(p; s_k)} \\ &= p^{\sum_{i>h} s_i s_h (e_i - e_h - 1)} g(p; n, s_1) g(p; n - s_1, s_2) \cdots g(p; n - s_1 - \cdots - s_{k-2}, s_{k-1})\end{aligned}\quad (2)$$

as an integral function of the indeterminate prime number p .

When several classes $\alpha, \beta, \dots$ are considered simultaneously, it is understood that the indeterminate prime number p is the same for all of them. Thus expressions such as " α is divisible by β " have an unambiguous meaning. Suppose that we have s classes

$$\alpha_i = \text{Cl}(a_{i1} \mid a_{i2} \mid \cdots \mid a_{in}) \quad (i = 1, \dots, s),$$

and let

$$\gamma' = [\alpha_1, \dots, \alpha_s] = \text{Cl}(c'_1 \mid \cdots \mid c'_n)$$

be their least common multiple and

$$\delta' = (\alpha_1, \dots, \alpha_s) = \text{Cl}(d'_1 \mid \cdots \mid d'_n)$$

their greatest common divisor. Then c'_k is the greatest, and d'_k the least, of the numbers $a_{1k}, \dots, a_{sk}$. The definition given in § 7 also immediately shows how to determine the limiting multiple

$$\gamma'' = [|\alpha_1, \dots, \alpha_s|]$$

and the limiting divisor

$$\delta'' = (|\alpha_1, \dots, \alpha_s|)$$

of the classes $\alpha_1, \dots, \alpha_s$: order the ns exponents of $\alpha_1, \dots, \alpha_s$ by size, so that they form a non-decreasing sequence. If $d''_1, \dots, d''_n$ are the n smallest and $c''_1, \dots, c''_n$ the n largest, then

$$\gamma'' = \text{Cl}(c''_1 \mid \cdots \mid c''_n), \quad \delta'' = \text{Cl}(d''_1 \mid \cdots \mid d''_n).$$

We express the fact that α is divisible by β by the inequality $\alpha \geq \beta$ or $\beta \leq \alpha$; the notations $\alpha > \beta$ and $\beta < \alpha$ exclude equality.

If $\alpha \geq \beta$ is not satisfied, then $t(\alpha \mid \beta)$ and $v(\alpha \mid \beta)$ vanish identically, i.e. for every p . Otherwise their meaning is given by § 8, III, IV, and the functions $\bar{\gamma}(\gamma; \alpha_1, \dots, \alpha_s)$ and $\bar{\delta}(\delta; \alpha_1, \dots, \alpha_s)$ are similarly given by § 9, Ia, IIa. All these functions are, as is easy to see, integral functions of p . The presentation becomes clearer if we regard p as an arbitrarily variable real quantity. We indicate this externally by replacing the letter p by x .

2. Thus now

$$\psi(\alpha) = x^{\sum_{i>h} s_i s_h (e_i - e_h)} \frac{\vartheta(x; n)}{\vartheta(x; s_1) \cdots \vartheta(x; s_k)} = x^{\sum_{i>h} s_i s_h (e_i - e_h - 1)} g(x; n, s_1) \cdots g(x; n - s_1 - \cdots - s_{k-2}, s_{k-1}),$$

where $s_1, \dots, s_k, e_1, \dots, e_k$ have their earlier meanings. Thus, if the exponents of α are not numerically given, one must still know the numbers $s_1, \dots, s_k$ in order to form $\psi(\alpha)$; that is, one must know whether two successive exponents are equal or not. Even greater inconveniences arise for t and v , and the treatment of $\bar{\gamma}$ and $\bar{\delta}$ becomes completely opaque. These difficulties can be removed by replacing the exponents by other quantities, which we introduce under the name "indices".

By the "sequence of indices" of a class α we mean an infinite sequence of integers

$$a_1, a_2, a_3, \dots,$$

whose h th element a_h states how many exponents of α are smaller than h . For example, to the classes

$$\text{Cl}(2 \mid 3 \mid 3 \mid 5 \mid 8), \quad \text{Cl}(0 \mid 4 \mid 7)$$

there correspond the sequences

$$0, 0, 1, 3, 3, 4, 4, 4, 5, 5, \dots; \quad 1, 1, 1, 1, 2, 2, 2, 3, 3, \dots$$

The following assertions are easy to verify.

1) The sequence of indices of a class of degree n is an infinite non-decreasing sequence of non-negative integers which rises to the number n .

2) Conversely, every sequence having the properties stated in 1) is the sequence of indices of a certain class α of degree n .

3) By definition, the h th index of α gives the number of exponents $< h$.

4) Conversely, the h th exponent of α gives the number of indices $\leq h$.

The sequence of indices $a_1, a_2, \dots$ of a class may also be continued in the negative direction. One then sets $a_0 = 0$, $a_{-1} = 0$, $a_{-2} = 0, \dots$

We add to assertions 1)–4) the following, whose proof is also easy.

5) If α and β are classes of degree n , with indices a_h and b_h , then $\alpha \geq \beta$ holds if and only if $a_h \leq b_h$ for every h .

Let now $\alpha_1, \alpha_2, \dots, \alpha_s$ be classes of degree n , and put

$$[\alpha_1, \alpha_2, \dots, \alpha_s] = \gamma', \quad (\alpha_1, \alpha_2, \dots, \alpha_s) = \delta',$$

$$[|\alpha_1, \alpha_2, \dots, \alpha_s|] = \gamma'', \quad (|\alpha_1, \alpha_2, \dots, \alpha_s|) = \delta''.$$

Let $a_h^{(i)}$ ($i = 1, 2, \dots, s$), c'_h, d'_h, c''_h, d''_h denote the indices of the classes $\alpha_i, \gamma', \delta', \gamma'', \delta''$. From 5) it follows that:

6) The h th index c'_h of the least common multiple γ' of $\alpha_1, \dots, \alpha_s$ is the least, and the h th index d'_h of the greatest common divisor δ' is the greatest, among the h th indices $a_h^{(1)}, \dots, a_h^{(s)}$.

For the sequences of indices of the limiting multiple and limiting divisor one obtains:

7) The h th index c''_h of the limiting multiple γ'' is the greater of the two numbers

$$0 \quad \text{and} \quad n - \sum_{i=1}^s (n - a_h^{(i)}) = \left(\sum_{i=1}^s a_h^{(i)} \right) - (s-1)n,$$

and the h th index d''_h of their limiting divisor δ'' is the smaller of

$$n \quad \text{and} \quad \sum_{i=1}^s a_h^{(i)}.$$

Here $a_h^{(i)}$ denotes the indices of α_i .

From 4) one easily obtains:

8) Among other relations, the indices a_h and the exponents $\bar{a}_1, \dots, \bar{a}_n$ of a class α satisfy

$$(1-n)\bar{a}_1 + (3-n)\bar{a}_2 + (5-n)\bar{a}_3 + \dots + (n-1)\bar{a}_n = \sum_h a_h(n - a_h). \quad (3)$$

No bounds for the summation need be stated, since the relation is correct when h runs through all integers.

3. In § 6 the function $g(x; n, m)$ was introduced and defined for all integral values of n and m . We collect here several formulae and facts about this function which will be needed later.

- 1) $g(x; n, m) = \frac{(1-x^n)(1-x^{n-1}) \cdots (1-x^{n-m+1})}{(1-x)(1-x^2) \cdots (1-x^m)} \quad (m > 0),$
- 2) $g(x; n, 0) = 1,$
- 3) $g(x; n, m) = 0 \quad (m < 0),$
- 4) $g(x; 0, m) = 0 \quad (m \neq 0), \quad g(x; 0, 0) = 1,$
- 5) $g(x; n, m) = g(x; n-1, m) + x^{n-m}g(x; n-1, m-1),$
- 6) $g(x; n, m) = g(x; n-1, m-1) + x^m g(x; n-1, m),$
- 7) $g(x; n, m) = x^{m(n-m)}g(1/x; n, m),$
- 8) $g(x; s, q)g(x; s-q, r) = g(x; s, r)g(x; s-r, q),$
- 9) $g(x; -n, m) = (-1)^m x^{-mn - \frac{m(m-1)}{2}} g(x; n+m-1, m),$
- 10) $g(x; n, m) = g(x; n, n-m) \quad (n \geq 0),$
- 11) $g(1/x; n, m) = \frac{\vartheta(x; n)}{\vartheta(x; m)\vartheta(x; n-m)} \quad (n \geq m \geq 0).$

12) In all cases the expansion of g in powers of x contains only finitely many positive and negative powers, and the coefficients are non-negative integers.

13) For $n \geq 0$, $g(x; n, m)$ is an integral function of x ; for $n > 0$, $m \geq 0$, it does not vanish identically.

§ 11. The functions $\psi(\alpha)$, $t(\alpha \mid \beta)$, $v(\alpha \mid \beta)$ represented with the aid of the indices.

If the class

$$\alpha = \text{Cl}(\bar{a}_1 \mid \bar{a}_2 \mid \cdots \mid \bar{a}_n)$$

has indices a_h , then $a_h - a_{h-1}$ is the number of exponents equal to $h-1$. Taking account of $\vartheta(x; 0) = 1$, § 6, III gives

$$\psi(\alpha) = x^{(1-n)\bar{a}_1 + (3-n)\bar{a}_2 + \cdots + (n-1)\bar{a}_n} \frac{\vartheta(x; n)}{\prod_h \vartheta(x; a_h - a_{h-1})}. \quad (1)$$

But by § 10, 2, 8),

$$(1-n)\bar{a}_1 + (3-n)\bar{a}_2 + \cdots + (n-1)\bar{a}_n = \sum_h (n - a_h) a_h,$$

and

$$\begin{aligned} \frac{\vartheta(x; n)}{\prod_h \vartheta(x; a_h - a_{h-1})} &= \prod_h \frac{\vartheta(x; n - a_{h-1})}{\vartheta(x; n - a_h) \vartheta(x; a_h - a_{h-1})} \\ &= \prod_h g(1/x; n - a_{h-1}, n - a_h) \\ &= \prod_h g(1/x; a_h, a_{h-1}). \end{aligned}$$

Therefore, using § 10, 3, 7),

$$\begin{aligned} \text{I. } \psi(\alpha) &= x^{\sum_h (n - a_h) a_h} \prod_h g(1/x; n - a_{h-1}, n - a_h) \\ &= x^{\sum_h (n - a_h) a_h} \prod_h g(1/x; a_h, a_{h-1}) \end{aligned}$$

$$\begin{aligned}
&= x^{\sum_h (n-a_h)a_{h-1}} \prod_h g(x; n-a_{h-1}, n-a_h) \\
&= x^{\sum_h (n-a_h)a_{h-1}} \prod_h g(x; a_h, a_{h-1}).
\end{aligned}$$

Suppose that of the classes

$$\alpha = \text{Cl}(a'_1 \mid \cdots \mid a'_n), \quad \beta = \text{Cl}(b'_1 \mid \cdots \mid b'_n) = \beta_0,$$

the first is divisible by the second. Then (§ 8, III)

$$t(\alpha \mid \beta) = \frac{\psi(\beta)\psi(\beta_1)\cdots\psi(\beta_{n-2})}{\psi(\bar{\beta})\psi(\bar{\beta}_1)\cdots\psi(\bar{\beta}_{n-2})} = \frac{\psi(\beta)\psi(\beta_1)\cdots\psi(\beta_{n-1})}{\psi(\bar{\beta})\psi(\bar{\beta}_1)\cdots\psi(\bar{\beta}_{n-1})}. \quad (2)$$

The meaning of the classes $\bar{\beta} = \bar{\beta}_0, \beta_i, \bar{\beta}_i$ ($i = 1, \dots, n-1$) is given in § 8, (5). Denote the indices of the classes $\alpha, \beta = \beta_0, \bar{\beta} = \bar{\beta}_0, \beta_i, \bar{\beta}_i$ by

$$a_h, \quad b_h = b_h^{(0)}, \quad \bar{b}_h = \bar{b}_h^{(0)}, \quad b_h^{(i)}, \quad \bar{b}_h^{(i)} \quad (i = 1, \dots, n-1).$$

Since β_i and $\bar{\beta}_i$ ($i = 0, \dots, n-1$) have degree $n-i$, one obtains

$$t(\alpha \mid \beta) = x^{\sum_h \sum_{i=0}^{n-1} \{(n-i-b_h^{(i)})b_h^{(i)} - (n-i-\bar{b}_h^{(i)})\bar{b}_h^{(i)}\}} \prod_h \prod_{i=0}^{n-1} \frac{g(1/x; b_h^{(i)}, b_{h-1}^{(i)})}{g(1/x; \bar{b}_h^{(i)}, \bar{b}_{h-1}^{(i)})}. \quad (3)$$

We now transform this expression so that it contains only the indices of α and β .

For $a_h < i$, the exponent of α is smaller than h , and hence, by § 8, (5),

$$b_h^{(i)} = 0, \quad \bar{b}_h^{(i)} = 0.$$

For $a_h = i$, one has $a'_i < h \leq a'_{i+1}$, and therefore

$$\bar{b}_h^{(i)} = 0, \quad b_h^{(i)} = b_h - i = b_h - a_h.$$

For $a_h > i$, one has $a'_{i+1} < h$, and therefore

$$b_h^{(i)} = \bar{b}_h^{(i)} = b_h - i.$$

It follows that

$$\sum_h \sum_{i=0}^{n-1} \{(n-i-b_h^{(i)})b_h^{(i)} - (n-i-\bar{b}_h^{(i)})\bar{b}_h^{(i)}\} = \sum_h (n-b_h)(b_h - a_h), \quad (4)$$

and also

$$\prod_{i=0}^{n-1} \frac{g(1/x; b_h^{(i)}, b_{h-1}^{(i)})}{g(1/x; \bar{b}_h^{(i)}, \bar{b}_{h-1}^{(i)})} = g(1/x; b_h - a_{h-1}, b_{h-1} - a_{h-1}).$$

Consequently

$$\begin{aligned}
\text{II. } t(\alpha \mid \beta) &= x^{\sum_h (n-b_h)(b_h-a_h)} \prod_h g(1/x; b_h - a_{h-1}, b_{h-1} - a_{h-1}) \\
&= x^{\sum_h (n-b_h)(b_{h-1}-a_{h-1})} \prod_h g(x; b_h - a_{h-1}, b_{h-1} - a_{h-1}) \\
&= x^{\sum_h (n-b_h)(b_h-a_h)} \prod_h g(1/x; b_h - a_{h-1}, b_h - b_{h-1})
\end{aligned}$$

$$= x^{\sum_h (n-b_h)(b_{h-1}-a_{h-1})} \prod_h g(x; b_h - a_{h-1}, b_h - b_{h-1}).$$

By § 7, I,

$$v(\alpha | \beta) = \frac{\psi(\alpha)}{\psi(\beta)} t(\alpha | \beta).$$

Using the formulae for ψ , this gives

$$\begin{aligned} \text{III. } v(\alpha | \beta) &= x^{\sum_h (b_h - a_h) a_h} \prod_h g(1/x; b_h - a_{h-1}, b_h - a_h) \\ &= x^{\sum_h (b_h - a_h) a_{h-1}} \prod_h g(x; b_h - a_{h-1}, b_h - a_h) \\ &= x^{\sum_h (b_h - a_h) a_h} \prod_h g(1/x; b_h - a_{h-1}, a_h - a_{h-1}) \\ &= x^{\sum_h (b_h - a_h) a_{h-1}} \prod_h g(x; b_h - a_{h-1}, a_h - a_{h-1}). \end{aligned}$$

All these expressions were derived under the assumption $\alpha \geq \beta$. If that assumption is not fulfilled, one can choose h so that $b_{h-1} < a_{h-1} \leq b_h$; then the four expressions in II vanish identically, and the same holds for the four expressions in III. Since in this case in fact $t(\alpha | \beta) = 0$ and $v(\alpha | \beta) = 0$, the formulae hold without restriction.

§ 12. The degrees of the functions ψ , t , v , $\bar{\gamma}$, $\bar{\delta}$.

Since henceforth it will nowhere be necessary to consider classes of different degrees side by side, let the degree of all occurring classes be denoted by n .

The formulae § 11, I show that $\psi(\alpha)$ is an integral function of x with integral coefficients, that the coefficient of the highest power of x is 1, and that the degree of $\psi(\alpha)$ is

$$\sum_h (n - a_h) a_h, \quad (1)$$

where a_h are the indices of α . If β is a divisor of α , then (§ 11, II, III) $t(\alpha | \beta)$ and $v(\alpha | \beta)$ are likewise integral functions of x with integral coefficients, again with leading coefficient 1. Their degrees are respectively

$$\sum_h (n - b_h)(b_h - a_h), \quad (2)$$

$$\sum_h (b_h - a_h) a_h, \quad (3)$$

where b_h denotes the indices of β .

For $\bar{\gamma}(\gamma; \alpha_1, \dots, \alpha_s)$ and $\bar{\delta}(\delta; \alpha_1, \dots, \alpha_s)$ we have, by § 9, IIa, IIIa,

$$\begin{cases} \bar{\gamma}(\gamma; \alpha_1, \dots, \alpha_s) = t(\gamma | \alpha_1) \cdots t(\gamma | \alpha_s) - \sum_{\gamma' \leq \mu < \gamma} t(\gamma | \mu) \bar{\gamma}(\mu; \alpha_1, \dots, \alpha_s), \\ \bar{\delta}(\delta; \alpha_1, \dots, \alpha_s) = v(\alpha_1 | \delta) \cdots v(\alpha_s | \delta) - \sum_{\delta' \geq \mu > \delta} v(\mu | \delta) \bar{\delta}(\mu; \alpha_1, \dots, \alpha_s). \end{cases} \quad (4)$$

Here $\gamma' = [\alpha_1, \dots, \alpha_s]$, $\delta' = (\alpha_1, \dots, \alpha_s)$, and the summation is to be extended over all classes μ satisfying the conditions written under the summation signs. From (4) and the properties of t and v , one sees that $\bar{\gamma}$ and $\bar{\delta}$ are integral functions of x with integral coefficients, and also that $\bar{\gamma}$ vanishes when $\gamma \geq \gamma'$ is not satisfied and $\bar{\delta}$ vanishes when $\delta \leq \delta'$ is not satisfied.

We first examine the degree of $\bar{\delta}$ more closely and assume $\delta \leq \delta'$. The coefficient of the highest power of x cannot be negative in any $\bar{\delta}$ -function, since for a prime value of x the function counts a number. From (4), the degree of $\bar{\delta}(\delta; \alpha_1, \dots, \alpha_s)$ is therefore at most the degree of $v(\alpha_1 \mid \delta) \cdots v(\alpha_s \mid \delta)$, namely

$$\sum_{i=1}^s \sum_h (d_h - a_h^{(i)}) a_h^{(i)}, \quad (5)$$

where $a_h^{(i)}$ and d_h are the indices of α_i and δ . Only two cases are possible. Either the degrees of the functions

$$v(\mu \mid \delta) \bar{\delta}(\mu; \alpha_1, \dots, \alpha_s) \quad (6)$$

for $\delta' \geq \mu > \delta$ are all smaller than (5); then $\bar{\delta}$ reaches the stated maximal degree and the leading coefficient is 1. Or one of the functions (6) rises to degree (5), in which case the degree of $\bar{\delta}$ falls below this expression.

We prove that $\bar{\delta}(\delta; \alpha_1, \dots, \alpha_s)$ reaches the maximal degree (5) if and only if the class δ , already assumed to be a divisor of δ' , is a multiple of the limiting divisor δ'' of $\alpha_1, \dots, \alpha_s$. Let m_h denote the indices of the variable class μ . The maximum for the degree of

$$v(\mu \mid \delta) \bar{\delta}(\mu; \alpha_1, \dots, \alpha_s)$$

is

$$\begin{aligned} & \sum_h (d_h - m_h) m_h + \sum_{i=1}^s \sum_h (m_h - a_h^{(i)}) a_h^{(i)} \\ &= \sum_{i=1}^s \sum_h (d_h - a_h^{(i)}) a_h^{(i)} - \sum_h \left\{ (d_h - m_h) \left(\sum_{i=1}^s a_h^{(i)} - m_h \right) \right\}. \end{aligned} \quad (7)$$

Moreover (§ 10, 2, 7) and 5)) δ is a multiple of δ'' if and only if, for every h ,

$$d_h \leq \sum_{i=1}^s a_h^{(i)}. \quad (8)$$

If this inequality is always fulfilled, then from $\mu > \delta$ no term of

$$\sum_h \left\{ (d_h - m_h) \left(\sum_{i=1}^s a_h^{(i)} - m_h \right) \right\} \quad (9)$$

can be negative. At least one term must be positive, because equality of μ and δ is excluded. Hence the expression (7) is, for every μ , smaller than (5).

Conversely, if the conditions (8) are not all fulfilled, put

$$[\delta, \delta''] = \mu'.$$

The index m'_h of μ' is the smaller of $\sum_i a_h^{(i)}$ and d_h , and the class μ' occurs among the classes μ . By the preceding argument, $\bar{\delta}(\mu'; \alpha_1, \dots, \alpha_s)$ reaches the degree

$$\sum_{i=1}^s \sum_h (m'_h - a_h^{(i)}) a_h^{(i)},$$

and the function

$$v(\mu' \mid \delta) \bar{\delta}(\mu'; \alpha_1, \dots, \alpha_s)$$

rises to degree

$$\sum_{i=1}^s \sum_h (d_h - a_h^{(i)}) a_h^{(i)} - \sum_h \left\{ (d_h - m'_h) \left(\sum_{i=1}^s a_h^{(i)} - m'_h \right) \right\}.$$

Since the product

$$(d_h - m'_h) \left(\sum_{i=1}^s a_h^{(i)} - m'_h \right)$$

vanishes for every h , the degree of

$$v(\mu' \mid \delta) \bar{\delta}(\mu'; \alpha_1, \dots, \alpha_s)$$

is precisely (5).

In entirely similar fashion, for $\bar{\gamma}(\gamma; \alpha_1, \dots, \alpha_s)$ one obtains the following result. If γ is a multiple of $\gamma' = [\alpha_1, \dots, \alpha_s]$, then $\bar{\gamma}(\gamma; \alpha_1, \dots, \alpha_s)$ has degree at most

$$\sum_{i=1}^s \sum_h (n - a_h^{(i)}) (a_h^{(i)} - c_h), \quad (10)$$

where c_h denotes the indices of γ . The function reaches this degree if γ is a divisor of $\gamma'' = [[\alpha_1, \dots, \alpha_s]]$, and the leading coefficient is then 1; otherwise the maximal degree is not reached.

Since $\bar{\gamma}$ and $\bar{\delta}$ do not vanish identically in the cases where the stated maximal degree is reached, they must remain positive as soon as x exceeds a certain positive bound. It will later be shown that this already occurs for $x > 1$, so that the functions do not vanish for any prime value of x . It will also emerge that in all other cases, where so far it has only been shown that the maximal degree is not reached, the functions $\bar{\gamma}$ and $\bar{\delta}$ vanish identically.

This will complete the proof of Theorem V of § 7. In order to reach that point, however, it is not enough to consider the degrees of $\bar{\gamma}$ and $\bar{\delta}$. We must examine their nature more closely, first considering the functions from which, as will be shown later, $\bar{\gamma}$ and $\bar{\delta}$ can be constructed by multiplication, up to the insertion of a factor that is only a power of x , in the same way that ψ , t , and v were expressed by means of the g -functions.

§ 13. The ω -functions.

1. We denote by

$$\omega(x \mid k \mid e \mid q_1, r_1; q_2, r_2; \dots; q_s, r_s)$$

the sum

$$\sum_m (-1)^m x^{\frac{m(m-1)}{2} - km} g(x; e, m) g(x; q_1 - m, r_1) g(x; q_2 - m, r_2) \cdots g(x; q_s - m, r_s). \quad (1)$$

The quantity x is the argument, k the first parameter, e the second parameter, and q_1, r_1 form the first parameter pair, q_2, r_2 the second, and so on. All parameters are to be integers, and the sum extends over all integers m . If $e \geq 0$, to which case we may restrict ourselves here, only finitely many terms of the sum are non-zero. The ω -function is then a rational function of x , which can possibly become infinite only for $x = 0$ and $x = \infty$.

The function (1) is to be called a proper ω -function if no parameter is negative and moreover

$$q_i \geq \begin{cases} e, \\ r_i, \end{cases} \quad (i = 1, \dots, s)$$

holds. In this case

$$k + r_1 + \cdots + r_s - e$$

is called the critical number of the ω -function.

2. We first treat the ω -function without parameter pairs:

$$\omega(x \mid k \mid e) = \sum_m (-1)^m x^{\frac{m(m-1)}{2} - km} g(x; e, m).$$

We have

$$\omega(x \mid k \mid 0) = 1, \quad (2)$$

and for $e > 0$, using

$$g(x; e, m) = g(x; e-1, m-1) + x^m g(x; e-1, m),$$

one obtains

$$\begin{aligned} \omega(x \mid k \mid e) &= \sum_m (-1)^m x^{\frac{m(m-1)}{2} - km} g(x; e-1, m-1) \\ &\quad + \sum_m (-1)^m x^{\frac{m(m-1)}{2} - (k-1)m} g(x; e-1, m). \end{aligned} \quad (3)$$

Since

$$\frac{m(m-1)}{2} - km = \frac{(m-1)(m-2)}{2} - (k-1)(m-1) - k,$$

it follows that

$$\omega(x \mid k \mid e) = (1 - x^{-k}) \omega(x \mid k-1 \mid e-1), \quad (4)$$

and therefore, using (2),

$$\omega(x \mid k \mid e) = (1 - x^{-k})(1 - x^{-k+1}) \cdots (1 - x^{-k+e-1}). \quad (5)$$

If this ω -function is proper, so that $k \geq 0$ and $e \geq 0$, then $k - e$ is its critical number. From (5) one sees that the function vanishes identically when the critical number is negative, while otherwise it has a positive non-zero value for every $x > 1$.

3. For an ω -function with s parameter pairs, assuming only that its second parameter e is not negative, application of

$$g(x; q_s - m, r_s) = g(x; q_s - 1 - m, r_s - 1) + x^{r_s} g(x; q_s - 1 - m, r_s)$$

gives

$$\begin{aligned} \omega(x \mid k \mid e \mid q_1, r_1; \dots; q_s, r_s) \\ &= x^{r_s} \omega(x \mid k \mid e \mid q_1, r_1; \dots; q_{s-1}, r_{s-1}; q_s - 1, r_s) \\ &\quad + \omega(x \mid k \mid e \mid q_1, r_1; \dots; q_{s-1}, r_{s-1}; q_s - 1, r_s - 1), \end{aligned} \quad (6)$$

and hence, more generally,

$$\begin{aligned} \omega(x \mid k \mid e \mid q_1, r_1; \dots; q_s, r_s) \\ &= \sum_{l=0}^t C'_l \omega(x \mid k \mid e \mid q_1, r_1; \dots; q_{s-1}, r_{s-1}; q_s - t, r_s - l), \end{aligned} \quad (7)$$

where t is any non-negative integer. By induction one finds

$$C'_l = x^{l(r_s - t)} g(x; t, l),$$

and what matters here is that this coefficient is positive for every positive value of x .

4. In the special case $q_s = e$, with $e \geq 0$, the identity

$$g(x; e, m) g(x; e - m, r_s) = g(x; e, r_s) g(x; e - r_s, m)$$

gives

$$\begin{aligned} & \omega(x \mid k \mid e \mid q_1, r_1; \dots; q_{s-1}, r_{s-1}; e, r_s) \\ &= g(x; e, r_s) \omega(x \mid k \mid e - r_s \mid q_1, r_1; \dots; q_{s-1}, r_{s-1}). \end{aligned} \quad (8)$$

The ω -function on the right contains one fewer parameter pair. The factor $g(x; e, r_s)$ vanishes if $r_s < 0$ or $r_s > e$; otherwise, for positive x , it has a positive value.

5. Let now

$$\omega(x \mid k \mid e \mid q_1, r_1; \dots; q_s, r_s)$$

be a proper ω -function. Then $q_s - e \geq 0$, and applying (7) with $t = q_s - e$ and then (8) gives

$$\begin{aligned} & \omega(x \mid k \mid e \mid q_1, r_1; \dots; q_s, r_s) \\ &= \sum_{\substack{r_s + e - q_s \leq h \leq r_s \\ 0 \leq h \leq e}} C_h \omega(x \mid k \mid e - h \mid q_1, r_1; \dots; q_{s-1}, r_{s-1}). \end{aligned} \quad (10)$$

The coefficients C_h are positive for every positive x . The ω -functions on the right are proper and contain only $s - 1$ parameter pairs. Thus every proper ω -function can be expressed linearly and homogeneously in proper ω -functions without parameter pairs, with coefficients positive for positive x .

The critical number of the function considered is $k + r_1 + \dots + r_s - e$, while that of

$$\omega(x \mid k \mid e - h \mid q_1, r_1; \dots; q_{s-1}, r_{s-1})$$

is

$$k + r_1 + \dots + r_{s-1} + h - e,$$

which is never larger than $k + r_1 + \dots + r_s - e$ for the values of h under consideration. Hence a proper ω -function with negative critical number is a homogeneous linear combination of such functions without parameter pairs and therefore vanishes.

If the critical number $k + r_1 + \dots + r_s - e$ is non-negative, then at least one term on the right of (10) remains whose critical number is not negative, namely the term obtained by taking h equal to the smaller of e and r_s . Hence the given ω -function can be expressed as a homogeneous linear combination of proper ω -functions without parameter pairs of non-negative critical number, and at least one such function actually occurs. Since the coefficients are positive for positive x , and the ω -functions are positive for $x > 1$, the original ω -function is also positive for $x > 1$.

We have reached the important conclusion:

- I. A proper ω -function whose critical number is < 0 vanishes identically.
- II. A proper ω -function whose critical number is ≥ 0 has a positive value for $x > 1$.

§ 14. The functions $\bar{t}$ and $\bar{v}$.

For $e > 0$,

$$0 = \omega(x \mid 0 \mid e) = \sum_m (-1)^m x^{m(m-1)/2} g(x; e, m). \quad (1)$$

If a, b, c are integers satisfying

$$0 \leq c \leq a < b, \quad (2)$$

then replacing e by $b - a$ and m by $b - m$ in (1), multiplying by

$$x^{c(b-a)} g(x; b - c, a - c),$$

and using

$$\begin{aligned} g(x; b-a, b-m)g(x; b-c, a-c) &= g(x; m-c, a-c)g(x; b-c, b-m) \\ &= g(x; m-c, m-a)g(x; b-c, b-m), \end{aligned}$$

one obtains

$$\sum_m (-1)^{b-m} x^{(b-m)(b-m-1)/2+c(b-a)} g(x; m-c, m-a)g(x; b-c, b-m) = 0. \quad (3)$$

Let α and β be two classes with indices a_h, b_h . Put

$$\begin{aligned} F_h &= \sum_{m_h} (-1)^{b_h-m_h} x^{(b_h-m_h)(b_h-m_h-1)/2+(m_h-a_h)a_{h-1}+(b_h-m_h)b_{h-1}} \\ &\quad \cdot g(x; m_h-a_{h-1}, m_h-a_h)g(x; b_h-b_{h-1}, b_h-m_h). \end{aligned} \quad (4)$$

For each h this sum has only finitely many non-zero terms, and $F_h = 1$ if $h \leq 0$ or if h is larger than all exponents of α and β .

The summation may be restricted to values of m_h satisfying

$$b_h \geq m_h \geq \begin{cases} b_{h-1}, \\ a_h, \end{cases} \quad (5)$$

so F_h vanishes when $a_h > b_h$. It also vanishes when $a_{h-1} = b_{h-1}$ and at the same time $a_h < b_h$. In contrast, if $a_{h-1} = b_{h-1}$ and $a_h = b_h$, then $F_h = 1$. Consequently

$$\prod_h F_h$$

has value 1 or 0 according as α and β are equal or different. In the special case where the number s of classes appearing in $\bar{\gamma}(\gamma; \alpha_1, \dots, \alpha_s)$ or $\bar{\delta}(\delta; \alpha_1, \dots, \alpha_s)$ is reduced to 1, the definitions of § 7 give exactly the same alternatives. Thus

$$\prod_h F_h = \bar{\delta}(\alpha; \beta). \quad (6)$$

Expanding the product gives

$$\begin{aligned} \prod_h F_h &= \sum_{\dots, m_0, m_1, \dots} \left\{ (-1)^{\sum_h (b_h-m_h)} x^{\sum_h \frac{(b_h-m_h)(b_h-m_h-1)}{2} + \sum_h (b_h-m_h)b_{h-1}} \prod_h g(x; b_h-b_{h-1}, b_h-m_h) \right. \\ &\quad \cdot x^{\sum_h (m_h-a_h)a_{h-1}} \prod_h g(x; m_h-a_{h-1}, m_h-a_h) \left. \right\}. \end{aligned} \quad (7)$$

It suffices to sum over sequences m_h satisfying

$$b_h \geq m_h \geq a_h, \quad m_h \geq m_{h-1}, \quad (8)$$

or even over all sequences satisfying

$$m_h \geq m_{h-1} \geq 0, \quad \lim_{h \rightarrow \infty} m_h = n. \quad (9)$$

The latter conditions say exactly that the m_h are the indices of a class μ of degree n , while (8) also says that

$$\alpha \geq \mu \geq \beta.$$

Under this assumption the second factor in (7) is $v(\alpha \mid \mu)$, while the first defines a function of β and μ , denoted $\bar{v}(\mu \mid \beta)$:

$$\bar{v}(\mu \mid \beta) = (-1)^{\sum_h (b_h - m_h)} \prod_h x^{(b_h - m_h)(b_h - m_h - 1)/2 + (b_h - m_h)b_{h-1}} g(x; b_h - b_{h-1}, b_h - m_h). \quad (10)$$

Thus (6) becomes

$$\sum_{\mu} v(\alpha \mid \mu) \bar{v}(\mu \mid \beta) = \bar{\delta}(\alpha; \beta). \quad (11)$$

From this relation, by induction over the classes below α , one obtains the inverse relation

$$\sum_{\mu} \bar{v}(\alpha \mid \mu) v(\mu \mid \beta) = \bar{\delta}(\alpha; \beta). \quad (14)$$

Indeed $\bar{v}(\alpha \mid \beta)$ is determined by

$$\bar{v}(\alpha \mid \beta) = \bar{\delta}(\alpha; \beta) - \sum_{\mu < \alpha} v(\alpha \mid \mu) \bar{v}(\mu \mid \beta), \quad (13)$$

and the cases $\alpha \not\geq \beta$ and $\alpha = \beta$ give $\bar{v}(\alpha \mid \beta) = 0$ and 1, respectively. Subtracting the already known lower-class terms proves (14) in general.

For

$$\bar{t}(\mu \mid \beta) = \frac{\psi(\beta)}{\psi(\mu)} \bar{v}(\mu \mid \beta) \quad (16)$$

one obtains, using (11) and (14),

$$\sum_{\mu} t(\alpha \mid \mu) \bar{t}(\mu \mid \beta) = \bar{\gamma}(\alpha; \beta), \quad (17)$$

$$\sum_{\mu} \bar{t}(\alpha \mid \mu) t(\mu \mid \beta) = \bar{\gamma}(\alpha; \beta). \quad (18)$$

Direct calculation from (16) gives

$$\bar{t}(\mu \mid \beta) = (-1)^{\sum_h (b_h - m_h)} \prod_h x^{(b_h - m_h)(b_h - m_h - 1)/2 + (n - m_h)(b_{h-1} - m_{h-1})} g(x; m_h - m_{h-1}, b_{h-1} - m_{h-1}). \quad (19)$$

By § 9,

$$\sum_{\mu} t(\nu \mid \mu) \bar{\gamma}(\mu; \alpha_1, \dots, \alpha_s) = t(\nu \mid \alpha_1) t(\nu \mid \alpha_2) \cdots t(\nu \mid \alpha_s), \quad (20)$$

$$\sum_{\mu} v(\mu \mid \lambda) \bar{\delta}(\mu; \alpha_1, \dots, \alpha_s) = v(\alpha_1 \mid \lambda) v(\alpha_2 \mid \lambda) \cdots v(\alpha_s \mid \lambda). \quad (21)$$

Multiplying by $\bar{t}(\gamma \mid \nu)$ and $\bar{v}(\lambda \mid \delta)$ respectively, and summing over ν and λ , gives

$$\bar{\gamma}(\gamma; \alpha_1, \dots, \alpha_s) = \sum_{\nu} \bar{t}(\gamma \mid \nu) t(\nu \mid \alpha_1) \cdots t(\nu \mid \alpha_s), \quad (22)$$

$$\bar{\delta}(\delta; \alpha_1, \dots, \alpha_s) = \sum_{\lambda} \bar{v}(\lambda \mid \delta) v(\alpha_1 \mid \lambda) \cdots v(\alpha_s \mid \lambda). \quad (23)$$

§ 15. Representation of $\bar{\gamma}$ and $\bar{\delta}$ by ω -functions.

The last two equations lead directly to representations of $\bar{\gamma}(\gamma; \alpha_1, \dots, \alpha_s)$ and $\bar{\delta}(\delta; \alpha_1, \dots, \alpha_s)$ by ω -functions. Let $a_h^{(i)}, c_h, d_h$ denote the indices of α_i ($i = 1, \dots, s$), γ , and δ .

We discuss $\bar{\delta}$ somewhat more fully. Substitution of the formulae for v and $\bar{v}$ gives

$$\begin{aligned} & \bar{v}(\mu \mid \delta) v(\alpha_1 \mid \mu) \cdots v(\alpha_s \mid \mu) \\ &= \prod_h \left\{ (-1)^{d_h - m_h} x^{(d_h - m_h)(d_h - m_h - 1)/2 + (d_h - m_h)d_{h-1} + (m_h - a_h^{(1)})a_{h-1}^{(1)} + \cdots + (m_h - a_h^{(s)})a_{h-1}^{(s)}} \right. \\ & \quad \cdot g(x; d_h - d_{h-1}, d_h - m_h) g(x; m_h - a_{h-1}^{(1)}, a_h^{(1)} - a_{h-1}^{(1)}) \cdots g(x; m_h - a_{h-1}^{(s)}, a_h^{(s)} - a_{h-1}^{(s)}) \left. \right\}. \end{aligned} \quad (1)$$

Here the m_h are the indices of μ . To obtain $\bar{\delta}(\delta; \alpha_1, \dots, \alpha_s)$ one sums all such products over the possible sequences m_h . Since any sequence which is not the sequence of indices of a class of degree n gives a vanishing product, that restriction may be dropped. Thus

$$\begin{aligned} \bar{\delta}(\delta; \alpha_1, \dots, \alpha_s) &= \sum_{\mu} \bar{v}(\mu \mid \delta) v(\alpha_1 \mid \mu) \cdots v(\alpha_s \mid \mu) \\ &= \prod_h \left\{ \sum_{m_h} \left[(-1)^{d_h - m_h} x^{(d_h - m_h)(d_h - m_h - 1)/2 + (d_h - m_h)d_{h-1} + (m_h - a_h^{(1)})a_{h-1}^{(1)} + \cdots + (m_h - a_h^{(s)})a_{h-1}^{(s)}} \right. \right. \\ & \quad \cdot g(x; d_h - d_{h-1}, d_h - m_h) g(x; m_h - a_{h-1}^{(1)}, a_h^{(1)} - a_{h-1}^{(1)}) \cdots g(x; m_h - a_{h-1}^{(s)}, a_h^{(s)} - a_{h-1}^{(s)}) \left. \right] \left. \right\}. \end{aligned} \quad (2)$$

Putting

$$(d_h - a_h^{(1)})a_{h-1}^{(1)} + \cdots + (d_h - a_h^{(s)})a_{h-1}^{(s)} = f_h, \quad (3)$$

$$a_{h-1}^{(1)} + \cdots + a_{h-1}^{(s)} - d_{h-1} = k_{h-1}, \quad (4)$$

$$d_h - d_{h-1} = e_h, \quad (5)$$

$$d_h - a_{h-1}^{(i)} = q_h^{(i)}, \quad a_h^{(i)} - a_{h-1}^{(i)} = r_h^{(i)} \quad (i = 1, \dots, s), \quad (6)$$

the sum occurring in (2) becomes

$$x^{f_h} \omega(x \mid k_{h-1} \mid e_h \mid q_h^{(1)}, r_h^{(1)}; \dots; q_h^{(s)}, r_h^{(s)}),$$

and hence

$$\text{I.} \quad \bar{\delta}(\delta; \alpha_1, \dots, \alpha_s) = \prod_h x^{f_h} \omega(x \mid k_{h-1} \mid e_h \mid q_h^{(1)}, r_h^{(1)}; \dots; q_h^{(s)}, r_h^{(s)}).$$

Each ω -function here is

$$\omega_h = \omega(x \mid k_{h-1} \mid e_h \mid q_h^{(1)}, r_h^{(1)}; \dots; q_h^{(s)}, r_h^{(s)}). \quad (7)$$

Only finitely many factors differ from 1. From (5) and (6),

$$e_h \geq 0, \quad r_h^{(i)} \geq 0. \quad (8)$$

For $\bar{\delta}(\delta; \alpha_1, \dots, \alpha_s)$ not to vanish identically it is necessary and sufficient that no ω_h vanish. Assuming this, one obtains successively

$$q_h^{(i)} \geq r_h^{(i)} \quad (i = 1, \dots, s), \quad (9)$$

which is equivalent to

$$d_h \geq a_h^{(i)} \quad (i = 1, \dots, s), \quad (10)$$

and to

$$q_h^{(i)} \geq e_h \quad (i = 1, \dots, s). \quad (11)$$

Furthermore the critical number of ω_h is

$$k_{h-1} + r_h^{(1)} + \dots + r_h^{(s)} - e_h = k_h,$$

so non-vanishing requires

$$k_h \geq 0. \quad (12)$$

Conversely, (8)–(12) imply that each ω_h is a proper ω -function with non-negative critical number, and therefore $\bar{\delta}$ is not identically zero and is positive for every $x > 1$. Relation (12) is equivalent to

$$d_h \leq \sum_{i=1}^s a_h^{(i)}. \quad (13)$$

Thus the necessary and sufficient conditions for non-vanishing of

$$\bar{\delta}(\delta; \alpha_1, \dots, \alpha_s)$$

are (10) and (13): the first says that δ is a divisor of $\delta' = (\alpha_1, \dots, \alpha_s)$, and the second that δ is a multiple of $\delta'' = (|\alpha_1, \dots, \alpha_s|)$.

For $\bar{\gamma}(\gamma; \alpha_1, \dots, \alpha_s)$ one obtains similarly, by putting

$$\begin{aligned} (n - a_{h+1}^{(1)})(a_h^{(1)} - c_h) + \dots + (n - a_{h+1}^{(s)})(a_h^{(s)} - c_h) &= f_h, \\ (n - a_{h+1}^{(1)}) + \dots + (n - a_{h+1}^{(s)}) - (n - c_{h+1}) &= k_{h+1}, \\ c_{h+1} - c_h &= e_h, \\ a_{h+1}^{(i)} - c_h &= q_h^{(i)}, \quad a_{h+1}^{(i)} - a_h^{(i)} = r_h^{(i)} \quad (i = 1, \dots, s), \end{aligned}$$

the formula

$$\text{II.} \quad \bar{\gamma}(\gamma; \alpha_1, \dots, \alpha_s) = \prod_h x^{f_h} \omega(x \mid k_{h+1} \mid e_h \mid q_h^{(1)}, r_h^{(1)}; \dots; q_h^{(s)}, r_h^{(s)}).$$

For $\bar{\gamma}$ not to vanish, all these ω -functions must be proper with non-negative critical number. Then $\bar{\gamma}$ is positive for every $x > 1$. The necessary and sufficient conditions are

$$c_h \leq a_h^{(i)}, \quad c_h \geq n - \sum_{i=1}^s (n - a_h^{(i)}).$$

The first says that γ is a multiple of $\gamma' = [\alpha_1, \dots, \alpha_s]$, and the second that γ is a divisor of $\gamma'' = [|\alpha_1, \dots, \alpha_s|]$.

This completes the proof of Theorem V, § 7.

Charlottenburg, 9 April 1898.